Introduction to Cybersecurity

Ass. Prof. Mohammed Abdel Hamid

Faculty of Computers & Information

Luxor University

Computer Science Dept.

Lecture 2

Lecture Rules

Cybersecurity Basics

Lecture Outline

- **What is Cybersecurity?**
- **What is the CIA Triad in Cybersecurity?**
- **Elements of Cybersecurity.**
- **What is Cyberattack?**
- **Types of Cyberattacks.**
- **Objectives of Cyberattacks.**
- **Technical Breaches of Cyberattack.**

What is Cybersecurity?

- “The **protection** of **software**, **hardware**, and **data resources connected** and **stored** on the **Internet**”, from an individual to a **large** corporation, **everybody** is concerned about the security of their **online data**, **software**, and **information**.

- The **protection** of the **personal**, **financial data**, **commercial data**, **business-critical** information, **operational continuity**, **data integrity**, and **availability** of **online software** services fall in the **cybersecurity domain.**

What is Cybersecurity?

- **Cybersecurity** is the **standard practices** that involve the **people**, **technology**, and **processes** in an **organization**, in a **team**, or even in a **stand-alone environment** in which the **computers** with the valuable **data** are **connected** to the **Internet** or the **Intranet**.

What is the CIA Triad in Cybersecurity?

- The **CIA Triad** is an **information security model**, It guides an organization's efforts towards ensuring **data security.**

- The three principles: **confidentiality**, **integrity**, and **availability** which is also the full for **CIA** in **cybersecurity**, form the **cornerstone** of a security **infrastructure**.

- It is ideal to apply these principles to any **security program.**

What is the CIA Triad in Cybersecurity?

- **Confidentiality:** This term **covers** two related concepts:
  — **Data confidentiality**
  — **Privacy**
- **Integrity:** This term **covers** two related concepts:
  — **Data integrity**
  — **System integrity**
- **Availability:** Ensuring **timely** and **reliable access and** use of **information** and **service**.

What is the CIA Triad in Cybersecurity?

What is the CIA Triad in Cybersecurity?

CIA Triad Example

An **ATM** incorporates measures to cover the principles of the triad:

- The **two-factor authentication** (**debit card** with the **PIN code**) provides **confidentiality** before **authorizing** access to **sensitive data**.
- The ATM and **bank software** ensure data **integrity** by **maintaining** all **transfer** and **withdrawal** records **made** via the **ATM** in the user's **bank accounting**.
- The ATM provides **availability** as it is for **public use** and is **accessible** at **all times**.

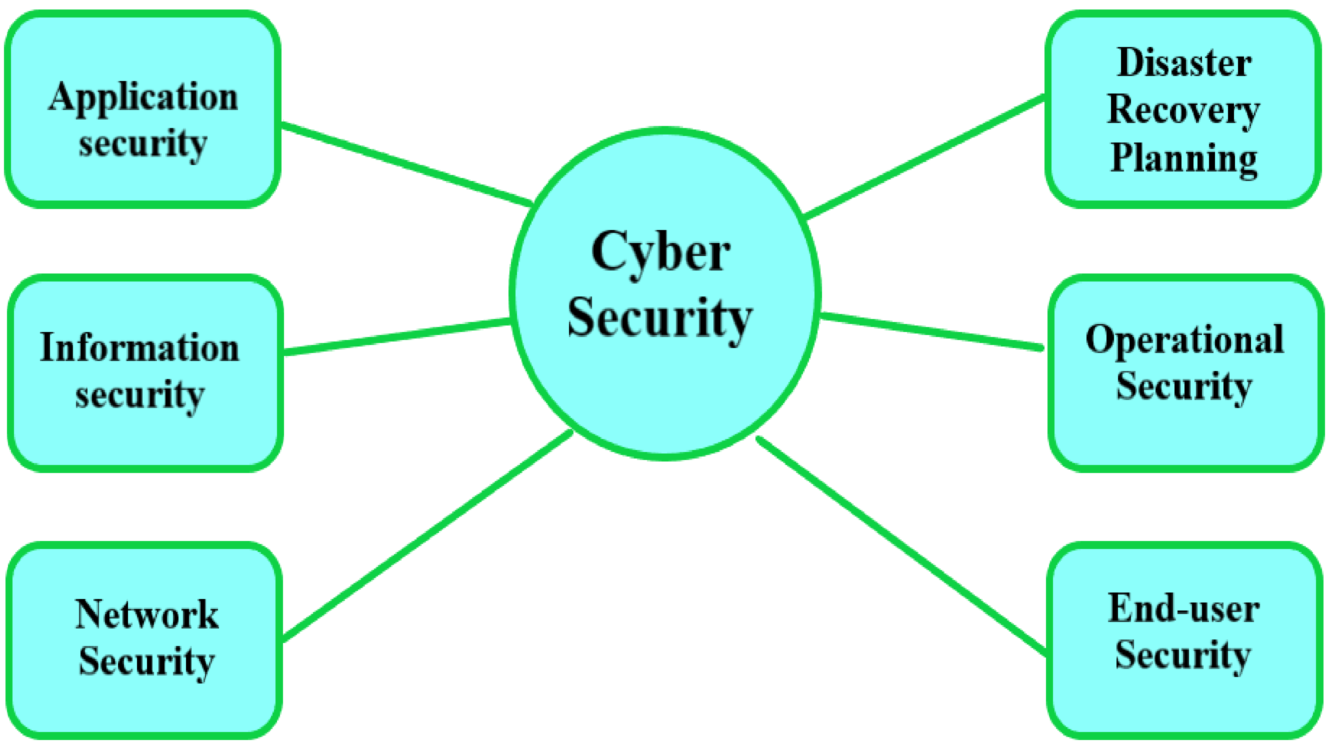
Elements of Cybersecurity

Cybersecurity can be classified into **multiple elements** as mentioned below:

- **Network security (NS).**
- **Information security (IS).**
- **Application security (AS).**
- **Disaster recovery planning \ Business continuity planning (BCP).**
- **Operational security (OPSEC).**
- **End-user security.**

Elements of Cybersecurity

- It is very **important** to note that **cybersecurity** is not just a **one-time** measure. It is a **continuous process** of security **awareness**, **strategic planning**, **implementation**, **monitoring**, and **evaluation**.



What is Cyberattack?

- **Malicious attempt** to gain **unauthorized** access to a **computer system**, **network**, or **digital device** with the **damage**, **steal**, or **manipulate data or other assets**. These attacks can target **individuals**, **organizations**, or even **governments**.
- **The main points (areas) of attacks (physical & logical targets):**
  - Data servers
  - Application servers
  - Storage servers
  - Financial information
  - Operational systems
  - Computer networks

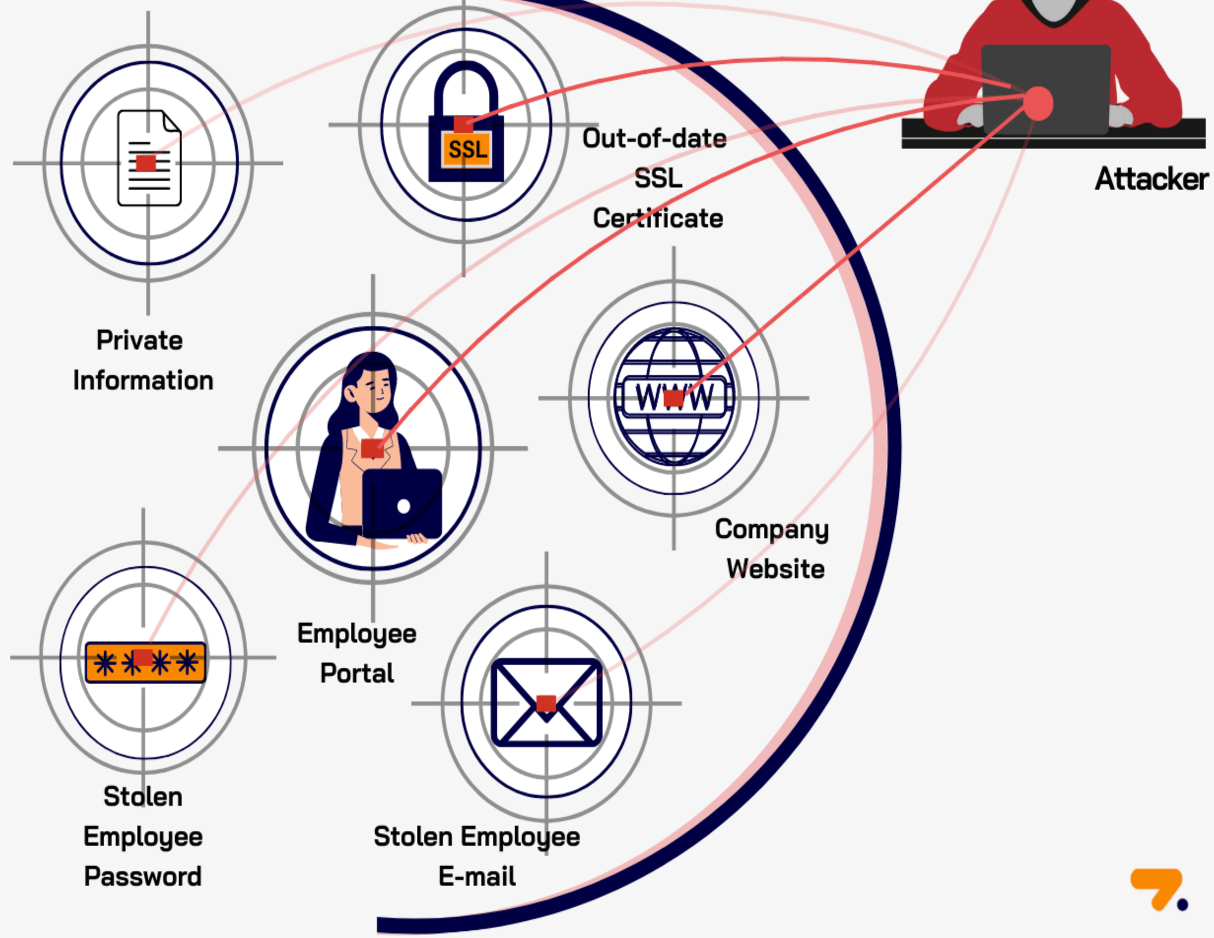
What is Cyberattack?

ATTACK SURFACE

Attack surface is known as the possible points where an unauthorized person can exploit the system with vulnerabilities. It's the combination of weak endpoints of software, system, or a network that attackers can penetrate.

Depending on various parameters and data types, attack surfaces are classified into three types:

1. **Digital** attack surface
2. **Physical** attack surface
3. **Social Engineering** Attack Surface

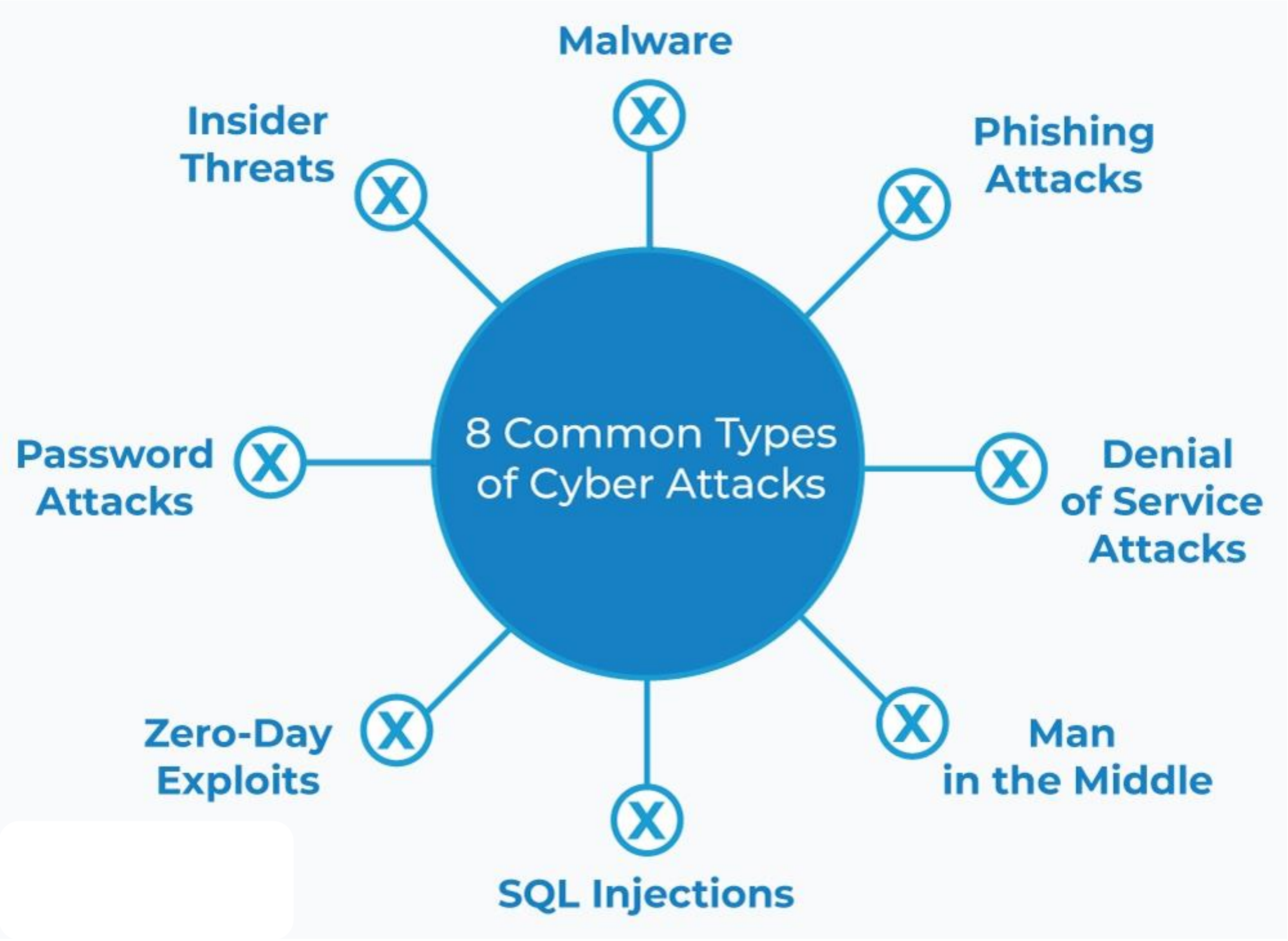


Types of Cyberattacks

- There are many types and methods used for **attacking** the **cyber environments** of the **other people**. A few important ones include the following:

    1. **Malware attacks.**
    2. **Phishing attacks.**
    3. **SQL Injection attacks.**
    4. **Denial-of-service** (**DoS**) **attacks**.
    5. **Man-in-the-middle** (**MITM**) **attacks.**
    6. **Password attacks.**
    7. **Zero-day Exploitation**
    8. **Insider attacks**

Types of Cyberattacks



Objectives of Cyberattacks

Some of the **main objectives** of a **hacker** for **conducting Cyberattacks** are listed below:

- **Achieving financial gains.**
- **Damaging** the **brand value** of the other party.
- **Inflicting damages** through cyberterrorism.
- **Obtaining government** and **business secrets**.
- **War Cyberattacks**.
- **Growth hacking email** campaign.

Objectives of Cyberattacks

Technical Breaches of Cyberattack

- **Confidentiality Breach.**

- **Integrity Breach.**

- **Availability Breach.**

Confidentiality Breach

- The **hackers** achieve the **confidentiality breach** through **multiple ways** as mentioned:
  - **Leaving computers with confidential information** without **monitoring**.
  - **Providing unauthorized access** from the **third-party person.**
  - **Unauthorized access** by **hackers** through **malware programs**.

Confidentiality Breach

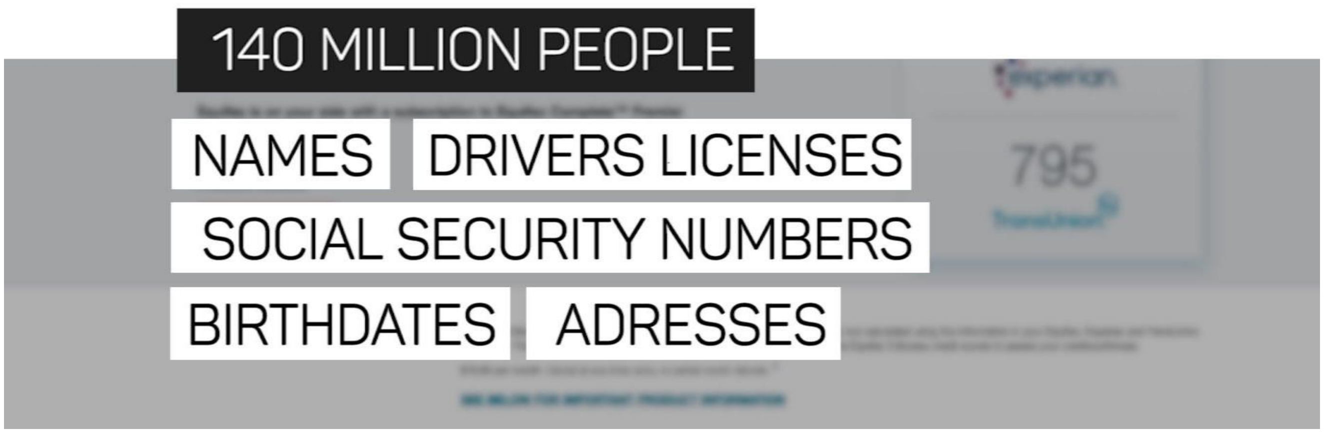
- **Examples** of **big confidential data breaches** in **recent years:**
  - **Marriot data breach (500 million person) in 2018**.
  - **Equifax data breach (143 million person) in 2017**.

Confidentiality Breach

- Hackers gained access to the personal information of **500 million** guests of Marriott International's Starwood.
- The information included **email addresses**, **passport numbers** **account information**, **date of birth**, **gender**, **reservation date**, **arrival** and **departure information.**

Confidentiality Breach

EQUIFAX®



- On **September 7, 2017, Equifax** announced that it had **breached** the **data** of approximately **143 million U.S. consumers.** The information included **Names, SSN, Birthdates, Addresses and driver Licenses.**

Integrity Breach

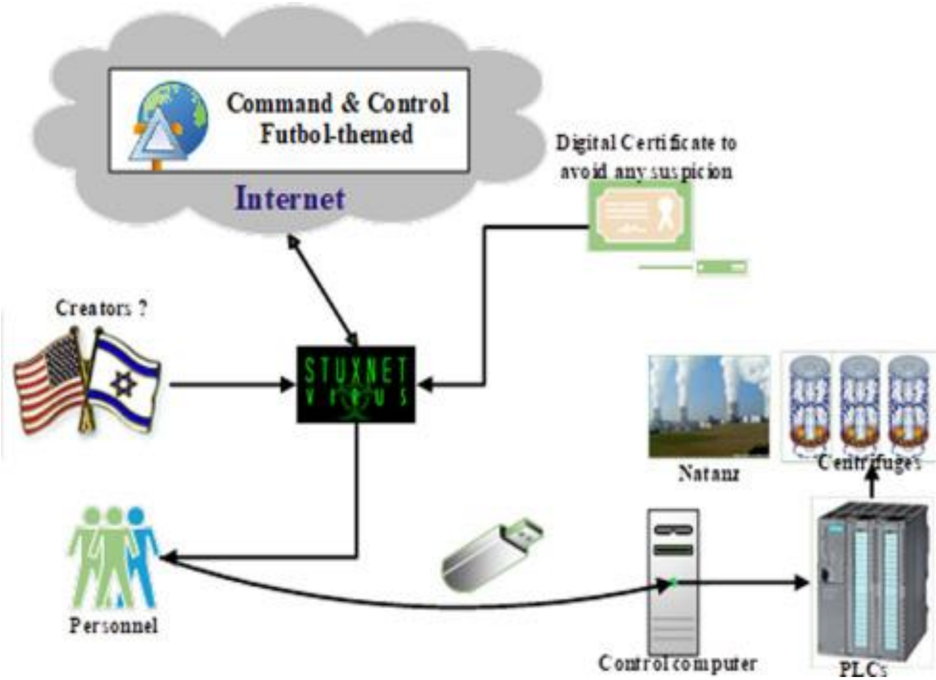
- The main sources of **integrity breach** may include the following:

  - **Installing malware** on the **server**.
  - **Malicious encryption** of data is undoable (**not retrieved**).
  - **Manipulation** (modification) of **original data**.
  - **Installation** of **viruses** on **hosts**.
  - **Insiders** with **malicious objectives**.

Integrity Breach

- **Examples** of **data integrity attacks:**
  - **Stuxnet** worm to manipulate the **Iranian Nuclear Program Data in 2010.**
  - **World Anti-Doping Agency** data manipulation **in 2016.**

Integrity Breach



- **Stuxnet**, a **computer worm** discovered in June 2010, is believed to have been used by the **United States and Israel** to attack **Iranian nuclear facilities.**
- Stuxnet initially spreads via **Windows,** **targeting Siemens** **industrial software and equipment**. It was **the first time** they had discovered **malware spying** on and **control industrial systems** using programmable logic controller **(PLC) rootkit.**

Integrity Breach

- **The World Anti-Doping Agency (WADA) confirmed in September 2016** that it had been attacked by **Russian** threat actors in a data leak **of** confidential information and modification about **forty-one athletes who competed in the Rio Olympic Games.**

Availability Breach

- The main sources of availability breach may include the following:

  - **Failure** of **hardware**.
  - **Error** (Malfunction) of **software**.
  - **Choking** (Limited) of **data bandwidth**.
  - **DoS attacks**.

Availability Breach

- **Examples** of breach of **availability** of services and data
  - **Failure** of **Google Cloud** in **February 2018.**
  - **Failure** of **Equinix** in **March 2018**.

Availability Breach

- A **database error** affecting **Google's application development platform** caused for some **high-profile Google Cloud customers** make **all services unavailable.**

Availability Breach

- **Equinix**, Inc. **connects enterprises** and **service providers** directly to **their customers** and **partners across** the **world's** most interconnected **data center** in the **Americas, Asia-Pacific, Europe, Middle East and Africa.**

Questions

Q1. Define output device of a computer?

Q2. What are storage components? Name some of them.

Q3. What are the recent technologies that could govern the future computers?

Q4. What are the basic elements of a computer network?

Q5. What are the major sources of breach of availability?

Q6. What is the CIA triad in Cybersecurity?

Answers

A1: A component that is connected to the computer for producing **meaningful information** processed from the raw data through input devices.

Examples: monitor, speaker… etc.

A2: The data storage bank to keep the data saved on the secondary location, where the data can be easily accessed and managed.

The storage components include the following:

- Hard disk drive (HDD)
- Flash disk drive (FDD)
- Optical disk drive (ODD)
- Magnetic tapes

Answers

A3: AI, quantum computing, nanotechnology, and parallel processing.

A4:

- Hosts
- Routers
- Switches
- Hubs
- Firewalls
- Servers
- Cables/links
- Protocols
- Connectors

Answers

A5: The main sources of breach of availability may include the following:

- Failure of hardware
- Malfunction of software
- Choking of data bandwidth
- Denial-of-service (DoS) attacks

Answers

A6: CIA Triad is an information security model, It guides an organization's efforts towards ensuring data security include the following:

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thanks for listening!